



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO DR MARAGATHAVALI & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

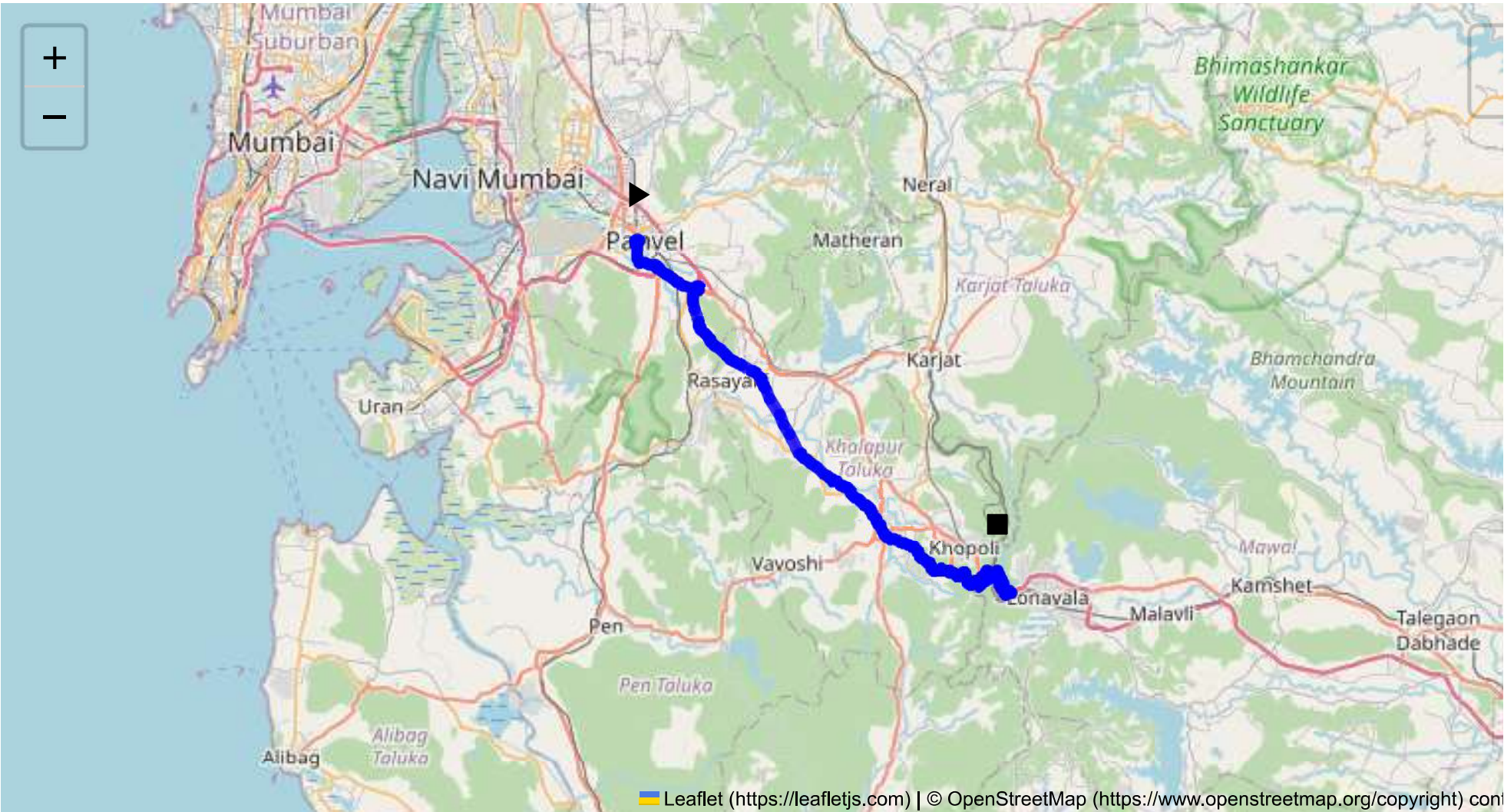
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.77 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (18.990713, 73.116844)
End: (18.7732, 73.3657)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report: Panvel to Tiger Valley, Lonavala

1. Overview of the Route Map

- The route begins at Panvel S.T. Bus Depot, utilizing the Mumbai-Pune Expressway to reach Tiger Valley in Khandala, Lonavala. This is a major highway predominantly throughout the journey with high-speed limits and generally well-maintained infrastructure.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- The region experiences a tropical climate. Monsoon season (June to September) brings heavy rainfall, increasing the risk of landslides and flooding, particularly affecting the expressway as it winds through hilly terrain.
- Fog can be an issue during winter months (December-January), reducing visibility and necessitating caution.

3. Analysis of Traffic Patterns

- Peak traffic typically occurs during morning (7:00 AM - 10:00 AM) and evening commutes (5:00 PM - 8:00 PM), particularly near Panvel junctions and toll plazas.
- Congestion is frequent near entry and exit points of the expressway and at interchanges around Lonavala.

4. Assessment of Road Quality and Infrastructure

- The Mumbai-Pune Expressway is a six-lane concrete road and generally well-maintained, but susceptible to potholes during and after the monsoon season.
- Lighting is adequate in most areas, but reduced on certain stretches, especially near exits and service roads.

5. Suggestions for Alternative Routes for Emergencies

- In cases of major blockages or emergencies, the old Mumbai-Pune Highway (NH4) serves as an alternate route, although it is narrower and slower due to mixed traffic and commercial setups.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Maharashtra state mandates specific timings for transporting hazardous materials, generally prohibiting transport during peak traffic hours.
- Necessary permits and documentation must be adhered to and kept ready for inspection at checkpoints.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- The Mumbai-Pune Expressway has a history of accidents involving heavy vehicles, often due to brake failures on steep gradients or driver fatigue. These incidents sometimes include spillage of hazardous materials, emphasizing the need for stringent adherence to safety protocols.

8. Environmental Considerations and Sensitive Areas

- The stretch between Lonavala and Khandala is environmentally sensitive, being part of the Western Ghats. Drivers are urged to minimize noise and pollution.
- There are animal crossing zones and frequent warnings of wildlife presence, necessitating vigilance.

9. Analysis of Communication Coverage

- The route is generally well-covered by all major mobile networks, though there may be short dead zones in the ghat sections near Khandala and Lonavala.

10. Estimated Emergency Response Times for Different Route Segments

- Emergency services are relatively accessible, with patrols on the expressway, but response times can vary from 15 to 30 minutes depending on the location and time of day.

11. Overall Summary of Risk Assessment

- The route is generally safe under normal weather conditions, with well-maintained infrastructure. However, significant risks include weather impacts on driving conditions, traffic congestion during peak times, and potential accidents involving heavy vehicles on steep or curved stretches.
- Emphasis on regular vehicle maintenance, adherence to speed limits, rest periods for drivers, and preparation for adverse weather conditions are critical for ensuring safety.
- Continuous monitoring of weather updates, traffic advisories, and adhering to local transport regulations will mitigate potential hazards associated with transporting hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Blind Spot	Blind Spot	18.98942, 73.11755	10 KM/Hr
1	Turn	High	18.98942, 73.11594	15 KM/Hr
2	Turn	Medium	18.96687, 73.13748	30 KM/Hr
3	Turn	Medium	18.95748, 73.15951	30 KM/Hr
4	Turn	Medium	18.95801, 73.15977	30 KM/Hr
5	Turn	Medium	18.95844, 73.15973	30 KM/Hr
6	Turn	Medium	18.95890, 73.15941	30 KM/Hr
7	Turn	Medium	18.76840, 73.33973	30 KM/Hr
8	Turn	Medium	18.76834, 73.34014	30 KM/Hr
9	Turn	Medium	18.77090, 73.34319	30 KM/Hr
10	Turn	Medium	18.76290, 73.34775	30 KM/Hr
11	Turn	Medium	18.76283, 73.34801	30 KM/Hr
12	Turn	Medium	18.76404, 73.35015	30 KM/Hr
13	Turn	Medium	18.76576, 73.35126	30 KM/Hr
14	Turn	Medium	18.76579, 73.35152	30 KM/Hr
15	Turn	Medium	18.76557, 73.35188	30 KM/Hr
16	Turn	Medium	18.76474, 73.35305	30 KM/Hr
17	Turn	Medium	18.77183, 73.35946	30 KM/Hr
18	Turn	Medium	18.77205, 73.36002	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
19	Turn	Medium	18.77191, 73.36046	30 KM/Hr
20	Turn	Medium	18.77128, 73.36078	30 KM/Hr
21	Turn	Medium	18.76286, 73.35397	30 KM/Hr
22	Turn	Medium	18.76282, 73.35513	30 KM/Hr
23	Turn	Medium	18.76684, 73.35964	30 KM/Hr
24	Turn	Medium	18.76675, 73.36015	30 KM/Hr
25	Turn	Medium	18.75849, 73.37218	30 KM/Hr
26	Turn	Medium	18.75827, 73.37250	30 KM/Hr
27	Turn	High	18.75882, 73.37505	15 KM/Hr
28	Turn	Medium	18.75862, 73.37518	30 KM/Hr
29	Blind Spot	Blind Spot	18.75837, 73.37578	10 KM/Hr
30	Turn	Medium	18.75740, 73.37190	30 KM/Hr
31	Turn	Medium	18.75759, 73.37164	30 KM/Hr
32	Turn	High	18.77072, 73.36643	15 KM/Hr
33	Turn	High	18.77096, 73.36654	15 KM/Hr

Risk Assessment - Elevation Changes

	Risk Type	Specific Risk	Risk Level	Coordinates	Speed Limit
0	Elevation Change	Ascent	High	18.91945915354813, 73.17259525532275	15 KMS/Hr
1	Elevation Change	Descent	High	18.9134082317682, 73.17940395140523	15 KMS/Hr
2	Elevation Change	Ascent	Medium	18.77198511901875, 73.33258782596738	30 KMS/Hr
3	Elevation Change	Ascent	Medium	18.76866771474596, 73.3406649178421	30 KMS/Hr
4	Elevation Change	Ascent	High	18.76651925285709, 73.3456575082452	15 KMS/Hr
5	Elevation Change	Ascent	High	18.76340116839086, 73.35552493762971	15 KMS/Hr
6	Elevation Change	Ascent	High	18.77121171710165, 73.36540738906407	15 KMS/Hr
7	Elevation Change	Descent	Medium	18.76828124292184, 73.36840845794214	30 KMS/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Dhamke Hospital	18.9923416, 73.1151144	30 km/h	Medium
1	hospital	Gune Hospital	18.9902872, 73.1144723	30 km/h	Medium
2	hospital	Siddhivinayak Hospital	18.9919712, 73.1146735	30 km/h	Medium
4	hospital	Gandhi Hospital	18.987502, 73.1153907	30 km/h	Medium
5	hospital	Dr.Prabhakar Patwardhan Smruti Rugnalaya	18.9888796, 73.1130896	30 km/h	Medium
6	hospital	Mhatre Accident Hospital	18.9913786, 73.1138808	30 km/h	Medium
7	hospital	Hande Hospital	18.989247, 73.1139461	30 km/h	Medium
9	clinic	Neo Clinic	18.9872731, 73.113066	30 km/h	Medium
11	clinic	Purohit Clinic	18.9849636, 73.1147861	30 km/h	Medium
12	clinic	Ameya Clinic	18.9855723, 73.1132117	30 km/h	Medium
13	clinic	Anand Mangal Maternity And Eye Clinic	18.9843178, 73.1135842	30 km/h	Medium
14	hospital	Likhite Orthopaedic & Paediatric Super Speciality Clinic& Nicu	18.9812237, 73.1144026	30 km/h	Medium
15	hospital	Pritam Hospital, Raigarh	18.9818971, 73.1141545	30 km/h	Medium
16	hospital	Sukam Hospital	18.9808357, 73.1152998	30 km/h	Medium
17	clinic	Siddhi Clinic	18.9797539, 73.1175072	30 km/h	Medium
18	hospital	ONGC Hospital	18.973353, 73.126321	30 km/h	Medium
19	hospital	Shifa Hospital	18.9724141, 73.1292526	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
3	college	Government College Of Education Panvel	18.9886481, 73.1169915	30 km/h	Medium
8	school	Gujrati school Ground	18.9885004, 73.1136858	30 km/h	Medium
10	school	Vithoba Khandappa School	18.9918839, 73.1129294	30 km/h	Medium
20	motorway_junction	Palaspa Junction	18.96845, 73.1328778	30 km/h	Medium
21	school	Swami Vivekananda Societies School	18.9568472, 73.1615144	30 km/h	Medium
22	college	National Institute of Securities Markets	18.9060834, 73.187517	30 km/h	Medium
23	motorway_junction	Khalapur	18.7996882, 73.2860473	30 km/h	Medium
24	motorway_junction	Khopoli	18.7963261, 73.2884961	30 km/h	Medium
25	motorway_junction	Khopoli Exit	18.7682065, 73.3394363	30 km/h	Medium
26	motorway_junction	Khandala Exit	18.7584876, 73.3733965	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 18.98942, 73.11755



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 18.98942, 73.11594



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 18.96687, 73.13748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.95748, 73.15951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.95801, 73.15977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.95844, 73.15973



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.95890, 73.15941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76840, 73.33973



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76834, 73.34014



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76404, 73.35015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76576, 73.35126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76579, 73.35152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76557, 73.35188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76474, 73.35305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.77183, 73.35946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.77205, 73.36002



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.77191, 73.36046



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.77128, 73.36078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76286, 73.35397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76282, 73.35513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76684, 73.35964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.76675, 73.36015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.75849, 73.37218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.75827, 73.37250



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 18.75882, 73.37505



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.75862, 73.37518



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 18.75837, 73.37578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.75740, 73.37190



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 18.75759, 73.37164



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 18.77072, 73.36643

